Top-K

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1. What *Top-K* Means

Top-K sampling is a decoding strategy that **limits the candidate pool** for the next token to the *K* tokens with the highest probabilities.

- **Smaller K** → **conservative** (model chooses only from the very top of the distribution).
- **Larger K** → **adventurous** (model may pick from a wide range of plausible tokens).

2. How It Works Under the Hood

1. The model assigns a probability to every possible next token.
2. These probabilities are sorted from highest to lowest.
3. Only the first *K* tokens are kept; everything else is discarded.
4. A single token is sampled from this truncated list (often combined with temperature).

   • **K = 1:** Equivalent to greedy decoding—always chooses the highest-probability token.

   • **K ↓:** Distribution becomes *tighter* → predictability rises.

   • **K ↑:** Distribution becomes *broader* → diversity rises.

3. Typical K Bands

Band	Output Character	When to Use
1 – 10	Very conservative, almost deterministic, low risk of	Technical explanations, code completion, compliance-critical

	hallucination	text
20 – 50	Balanced creativity and coherence	Chat agents, product copy, blog intros
50 – 100+	Highly diverse, playful, higher risk of drift	Poetry, brainstorming, storytelling, marketing slogans

> Rule of thumb: Lower K for precision, higher K for exploration.

4. Choosing the Right K

4.1 Precision-Driven Tasks

- Step-by-step reasoning, legal text, regex generation
- Recommended **K = 1 – 5** (often paired with **temperature ≤ 0.3**)

4.2 Balanced Output

- Long-form explanations, Q&A bots
- Recommended **K = 20 – 40** with **temperature ≈ 0.5-0.7**

4.3 Creative or Divergent Ideation

- Fiction writing, idea lists, humor
- Recommended **K = 60+** and **temperature ≥ 0.8** (consider using **top-p = 1** to avoid extra truncation)

5. Best-Practice Tips

1. **Combine with temperature**: Top-K gates the token list; temperature shapes randomness *within* that list.
2. **Tune iteratively**: Start with a moderate K (e.g., 40) and adjust up or down based on output quality.
3. **Pair with nucleus sampling (top-p)**: When both are applied, the stricter filter wins—useful for fine-grained control.
4. **Batch generate**: For ideation, sample multiple outputs at high K, then cherry-pick.
5. **Match K to context length**: High K on short prompts can cause topic drift faster than on longer, guiding prompts.

6. Common Pitfalls

- **Over-filtering**: K that is too small can make the prose robotic and bland.
- **Ignoring task requirements**: High K on fact-sensitive tasks invites hallucinations.
- **Assuming K alone solves everything**: Prompt clarity, temperature, stop sequences, and system instructions still matter.

7. Quick Decision Flow

1. **Do you need strict correctness?** → *Yes* → Use **K ≤ 5**.
2. **Do you need a balanced, human-like tone?** → *Yes* → Try **K ≈ 30-40**.
3. **Are you aiming for wild creativity?** → *Yes* → Push **K ≥ 60** and sample multiple completions.

Key takeaway: *Top-K is your diversity gate.* A low K keeps the model on the safest path; a high K opens the door to novel, surprising language—just be ready to guide or edit the results.